PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	MTOR INHIBITORS AND RISK OF OVARIAN CYSTS: A SYSTEMATIC REVIEW AND META-ANALYSIS.
AUTHORS	Parazzini, Fabio; Gerli, Sandro; Favilli, Alessandro; Vignali, Michele; Ricci, Elena; Cipriani, Sonia; Chiaffarino, Francesca; Dell'acqua, Andrea; Harari, Sergio; Bianchi, Stefano

VERSION 1 – REVIEW

REVIEWER	Diaz-Cueto, Laura UMAE Hospital de Gineco Obstetricia No. 4, Unidad de Investigación
REVIEW RETURNED	13-Feb-2021

GENERAL COMMENTS	Comments to Authors The authors realized a systematic review and meta-analysis of mTOR inhibitors and the risk of ovarian cyst. After analyzing the studies included in the meta-analysis, the pooled incidence of ovarian cysts was 37.0%, a percentage in the range reported in individual studies and clinical trials ranging between 10% to 50%. The manuscript is based on a limited number of studies, and the authors establish that it could be affected by potential limitations or bias. As an overall, the manuscript has minor revisions. It could be accepted if the authors make the following corrections and clarifications. 1. Please include in the introduction the mTOR inhibitors' clinical uses and focus on the use as an immunosuppressor. 2. In table 2, as a main characteristic, please include the ovaries study performed before treatment. 3. In table 2, sample size column, please be consistent with the nomenclature for all the rows (define use “= or :”). In the author column, the year is in parentheses or without parentheses. In the study design column, please check uppercase and lowercase. Also, include in each study or author the corresponding reference. 4. Page 8, line 24, It should be said “a total of seven studies have been identified” because in table 2 are seven studies. The authors previously clarified the study referenced as 13. Please, mark with an * the study excluded from the principal analysis with an annotation down the table. 5. Page 11, rows 3 to 10, be consistent with the study number (n=7), description of the type of study n=6, and description of women pathology included in the study n=7. In rows 19 to 26, please include the references. 6. Tabla 4, column standard treatment, second to the last row, please check if it should be said 8/469 and in the last row 8/487. 7. Page 18, Synthesis of results, please include the specific tables in the text.

	8. Please, finish the discussion considering the risk-benefit of mTOR inhibitors use as an immunosuppressor.

REVIEWER	Samartzis, Eleftherios University Hospital Zurich
REVIEW RETURNED	19-Feb-2021

GENERAL COMMENTS	This is a systematic review and meta-analysis assessing the risk of ovarian cysts in patients treated with the mTOR inhibitors sirolimus (or everolimus) as part of therapeutic immunosuppression schemes for allogenic renal or pancreatic islet transplantation. The systematic search identified 7 studies (4 renal and 3 pancreatic islet transplant studies) which were included in the quantitative synthesis. The research question is clearly defined following PICOS criteria and the PRISMA checklist has been added. It is somehow regrettable that the study protocol has not been prospectively registered. Otherwise, the methodology appears robust. The systematic review is informative and well written. However, as also mentioned by the authors, the clinical interpretability of the meta-analysis results and its significance is limited, mainly due to the very variable definitions and imaging modalities for ovarian cyst detection across the different studies. Major Points: 1) The authors should try to assess following points while respecting the given limits of their study protocol (no registered version of the protocol available): In the present form, the meta-analysis does not add a lot of value to the systematic review. As mentioned by the authors, the assessed outcomes (risk of ovarian cysts) across the studies were based on heterogeneous definitions of events. This includes variable imaging modalities and definitions (size, persistence) of an ovarian cyst. Whereas certain studies defined ovarian cysts as cystic formation >2cm in MRI images (Braun et al.), other studies included only cysts of >3cm not resolving spontaneously after 4 months diagnosed by transvaginal sonography (Cure et al). 2) Methods Table 1: PICOS criteria. Data extraction does not mention “clinical significant cysts” as a parameter. Please comment. 3) Data synthesis P7/L21-26 and Table 4 : The definition of “clinically significant cysts” also varies across the studies. Based on the performed analysis, the authors should clearly define the attributes making an ovarian cyst to be called “clinically significant” in their study. 4) P14-16/Systematic Review: Available data about severity of ovarian cysts: It appears that the milder cases are predominant in the literature (e.g., in the study of Bachmann et al.). However, some severe cases have been described, including iliac and ureter compression (Bachmann et al.), lower limb edema development (Ignatjovic et al.). It would probably have been possible to extract more specific data about cyst severity? Please comment. 5) P18/L35-38: Please also report the pooled OR for “clinically significant ovarian cysts” (within the limits of point 1 and 3) or comment. 6) Figure 2/ Forest Plot: it would possibly have been relevant to assess “clinically significant ovarian cyst” incidence separately as well (within the limits of point 1 and 3)? Please comment. Minor Points: 1) P13/14, Table 4: Please add percentages. 2) P15 / L60: The term “complex” is not sufficiently clear in this

	context. Please include the examples of cysts which the authors termed complex in the original publication. 3) P19 / L35-36: Pelvic (i.e., transvaginal) sonography would be the preferred imaging modality to exclude ovarian cysts.

VERSION 1 – AUTHOR RESPONSE

Reviewer:1
Dr. Laura Diaz-Cueto, UMAE Hospital de Gineco Obstetricia No. 4

1. Please include in the introduction the mTOR inhibitors' clinical uses and focus on the use as an immunosuppressor.
We have included in the introduction section the following sentences: “mTOR inhibitors are used as targeted therapy for tumors (in particular renal carcinoma). Further mTOR inhibitors inhibit T-cell proliferation and proliferative responses induced by several cytokines, including Interleukin 1, Interleukin 2, Interleukin 3, Interleukin 4, Interleukin 6, Insulin-like growth factor, Platelet-derived growth factor, and Colony-stimulating factors and they have been used in combination therapy with corticosteroids and cyclosporine in patients who received kidney transplantation to prevent organ rejection, and in the treatment of rheumatoid arthritis [1].”

2.In table 2, as a main characteristic, please include the ovaries study performed before treatment.
We have added a column in Table 2 indicating the method of ovary study before treatment.

3. In table 2, sample size column, please be consistent with the nomenclature for all the rows (define use “= or :”). In the author column, the year is in parentheses or without parentheses. In the study design column, please check uppercase and lowercase. Also, include in each study or author the corresponding reference.
We have corrected Table 2 as you required

4. Page 8, line 24, It should be said “a total of seven studies have been identified” because in table 2 are seven studies. The authors previously clarified the study referenced as 13. Please, mark with an * the study excluded from the principal analysis with an annotation down the table.
We have corrected and updated the main text and footnotes of Table 2 as you required.

5. Page 11, rows 3 to 10, be consistent with the study number (n=7), description of the type of study n=6, and description of women pathology included in the study n=7. In rows 19 to 26, please include the references.
We have corrected and updated, as you required.

6. Table 4, column standard treatment, second to the last row, please check if it should be said 8/469 and in the last row 8/487.
We have corrected the number.

7. Page 18, Synthesis of results, please include the specific tables in the text.
We have reported in the text the specific tables.

8. Please, finish the discussion considering the risk-benefit of mTOR inhibitors use as an immunosuppressor.

We have added the following sentence: “Further the risk of ovarian cysts, together with the impact of mTOR inhibitors on glucose metabolism, risk of diabetes and other potential adverse effects should be included in the risk benefit balance of mTOR Inhibitor use as immunosuppressive agents.”
Our expertise is in gynecology, thus we have preferred not to include a more assertive statement.

Reviewer: 2
Dr. Eleftherios Samartzis, University Hospital Zurich

1. The authors should try to assess following points while respecting the given limits of their study protocol (no registered version of the protocol available): In the present form, the meta-analysis does not add a lot of value to the systematic review. As mentioned by the authors, the assessed outcomes (risk of ovarian cysts) across the studies were based on heterogeneous definitions of events. This includes variable imaging modalities and definitions (size, persistence) of an ovarian cyst. Whereas certain studies defined ovarian cysts as cystic formation >2cm in MRI images (Braun et al.), other studies included only cysts of >3cm not resolving spontaneously after 4 months diagnosed by transvaginal sonography (Cure et al).
We have modified and updated the Discussion section with the following sentences: “The ascertainment of ovarian cysts was performed with different methodologies, variable imaging modalities and definitions (size, persistence) of ovarian cyst. Whereas certain studies defined ovarian cysts as cystic formation >2cm in MRI images [11], other studies included only cysts of >3cm not resolving spontaneously after 4 months diagnosed by transvaginal sonography [12,13], and in two studies the method was not reported [14,15]. Thus, in order to reduce the heterogeneity in the definition of ovarian cysts among the considered studies we have also performed a meta-analysis of clinically significant ovarian cysts”. (see also point 2)

2) Methods Table 1: PICOS criteria. Data extraction does not mention “clinical significant cysts” as a parameter. Please comment.
We have considered in the general analysis all the reported cysts as indicated in the PICOS
We have further performed a separate analysis for clinically significant cysts. We have better specified these points in the Methods section. In particular, we have added a column in Table 2 with the definition of clinically significant ovarian cysts and we have specified that the definition “clinically significant ovarian cysts” includes symptomatic cysts, large cysts (>6cm) and cysts requiring surgery. This point has been better reported in the systematic review section.

3) Data synthesis P7/L21-26 and Table 4 : The definition of “clinically significant cysts” also varies across the studies. Based on the performed analysis, the authors should clearly define the attributes making an ovarian cyst to be called “clinically significant” in their study.
See above

4) P14-16/Systematic Review: Available data about severity of ovarian cysts: It appears that the milder cases are predominant in the literature (e.g., in the study of Bachmann et al.). However, some severe cases have been described, including iliac and ureter compression (Bachmann et al.), lower limb edema development (Ignatjovic et al.). It would probably have been possible to extract more specific data about cyst severity? Please comment.
As previously underlined we have better described in the systematic review section the severity of the cysts observed in the various study added a column in Table 2 and presented in the meta-analysis section an analysis based on only cysts requiring surgery.

5) P18/L35-38: Please also report the pooled OR for “clinically significant ovarian cysts” (within the limits of point 1 and 3) or comment.
We have included the following sentence: “the pooled odds ratio for clinically significant ovarian cysts was 5.56 (95% CI 2.34-14.67)”

6) Figure 2/ Forest Plot: it would possibly have been relevant to assess “clinically significant ovarian cyst” incidence separately as well (within the limits of point 1 and 3)?

We reported in the Synthesis of Results section the following sentence: “we pooled the incidence of clinically significant ovarian cysts in studies reporting this information [8–10,12]. The resulting estimate was 17.3% (95% CI 5.6-29.1), heterogeneity chi-square 15.5, p<0.001) (Figure 3).”, and we added Figure 3 – Forest Plot, as you required.

Minor Points:

1) P13/14, Table 4: Please add percentages.
We have added percentages in Table 4, as you required.

2) P15 / L60: The term “complex” is not sufficiently clear in this context. Please include the examples of cysts which the authors termed complex in the original publication.
We have specified with the following sentence: “All the cysts were benign, but eight were considered complex because of haemorrhage, hydrosalpinx, cyst’s size (>6 cm), spontaneous rupture, or need of surgery for resolution.”

3) P19 / L35-36: Pelvic (i.e., transvaginal) sonography would be the preferred imaging modality to exclude ovarian cysts.
We have added the sentence you suggested: “Pelvic (i.e., transvaginal) sonography would be the preferred imaging modality to exclude ovarian cysts.”